CS & IT ENGINEERING

Operating System

CPU Scheduling

Lecture - 04

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Recap of Previous Lecture

- **Topic** SRTF Scheduling
- **Topic** LJF & LRTF Scheduling
- **Topic** HRRN Algorithm
- **Topic** Priority based algorithm

Topics to be Covered

- **Topic** Round Robin Algorithm
- **Topic** Multilevel Queue Scheduling
- **Topic** Multilevel Feedback Queue Scheduling

Topic : Round Robin (RR)

Scheduling Criteria: Arrival time + Quantum

Type of Algorithm: Preemptive

Quantum or time slice $\Rightarrow$ max amount of time a process can run on CPU

Topic : Round Robin (RR)

$Q = 2$

Process	Arrival Time	Burst Time	CT	TAT	WT
P1	0	3	9	9	6
P2	0	6	17	17	11
P3	0	4	13	13	9
P4	0	5	18	18	13

P1	P2	P3	P4	P1	P2	P3	P4	P2	P4

0 2 4 6 8 9 11 13 15 17 18

No. of context switches $= 9$

	BT	no. of times Process runs
P1	3	$\lceil 3/2 \rceil = 2$
P2	6	$\lceil 6/2 \rceil = 3$
P3	4	$\lceil 4/2 \rceil = 2$
P4	5	$\lceil 5/2 \rceil = 3$
		Total $= 10$

$Q = 2$

$$\left(\sum_{i=1}^{n} \left\lceil \frac{BT_i}{Q} \right\rceil \right) - 1$$

no. of context switches $= 10 - 1$

$= 9$

Ques)

	AT	BT
P1	0	5
P2	1	6
P3	2	3
P4	4	4

$Q = 3$

at Time	Ready Queue
0	~~P1~~
3	~~P2~~, P3, P1
6	P3, P1, P4, P2

P1	P2	P3	P1	P4	P2	P4

0 3 6 9 11 14 17 18

Ques)

	AT	BT	
P1	0	5	2
P2	2	2	
P3	3	1	

$Q = 3$

P1	P2	P3	P1

0 3 5 6 8

Time	Ready Queue
0	~~P1~~
3	P2 , P3 , P1

added at the end

(Process which comming from CPU)

Topic : Round Robin (RR)

$Q = 2$

Process	Arrival Time	Burst Time
P1	0	6
P2	1	5
P3	2	4
P4	3	3
P5	4	2
P6	5	4

Time	Ready Queue
0	~~P1~~
2	~~P2~~, P3, P1
4	~~P3~~, P1, P4, P5, P2
6	P1, P4, P5, P2, P6, P3

P1	P2	P3	P1	P4	P5	P2	P6	P3	P1	P4	P2	P6

0 2 4 6 8 10 12 14 16 18 20 21 22 24

Topic : Round Robin (RR)

$Q = 3$

Process	Arrival Time	Burst Time
P1	5	6
P2	8	7
P3	3	8
P4	6	3
P5	2	2
P6	4	4

At time	Ready Queue
2	P5
4	~~P3~~, P6
7	~~P6~~, P1, P4, P3
10	P1, P4, P3, P2, P6

	P5	P3	P6	P1	P4	P3	P2	P6	P1	P3	P2	P2

0 2 4 7 10 13 16 19 22 23 26 28 31 32

Topic : Round Robin (RR)

Q = 4

Process	Arrival Time	Burst Time
P1	0	12
P2	0	5
P3	3	9
P4	5	6
P5	2	8
P6	4	2
P7	1	7

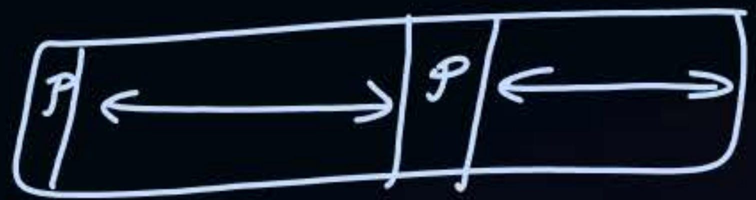
Time	Ready Queue
0	~~P1~~, P2
4	~~P2~~, P7, P5, P3, P6, P1
8	P7, P5, P3, P6, P1, P4, P2

P1	P2	P7	P5	P3	P6	P1	P4	P2	P7	P5	P3	P1	P4	P3

0 4 8 12 16 20 22 26 30 31 34 38 42 46 48 49

Topic : What Should Be the Quantum Value?

very-very small

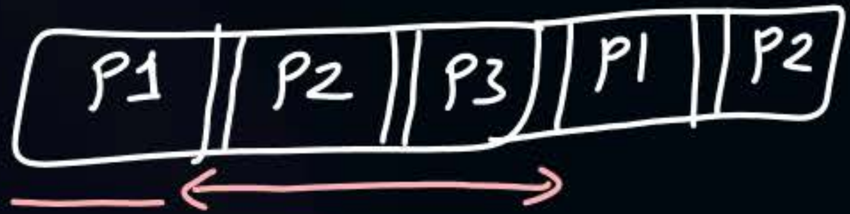


C.S. Time

CPU efficiency ≅ 0

small

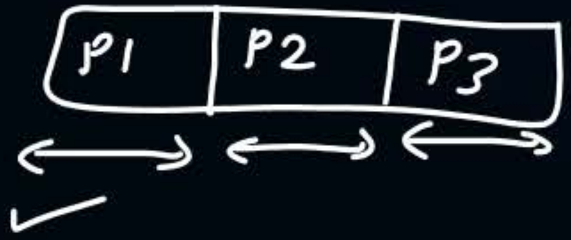
Interactive



large

Lesser Interactive

very-very large



RR algo may degrade to FCFS

Topic : RR

Advantages:

1. All processes execute one by one, so no starvation
2. Better interactiveness
3. Burst time is not required to be known in advance ⇒ RR Practicall implemented

Disadvantages:

1. Average waiting time and turnaround time ~~is~~ are more
2. Can degrade to FCFS

[MCQ]

#Q. If the time-slice used in the round-robin scheduling policy is more than the maximum time required to execute any process, then the policy will?

- **A** Degenerate to shortest job first
- **B** Degenerate to priority scheduling
- **C** Degenerate to first come first serve
- **D** None of the above

[MCQ]

H.W.

#Q. A scheduling algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (the lowest priority). The scheduler re-evaluates the process priorities every T time units and decides the next process to schedule. Which one of the following is TRUE if the processes have no I/O operations and all arrive at time zero?

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A This algorithm is equivalent to the first-come-first-serve algorithm

B This algorithm is equivalent to the round-robin algorithm.

C This algorithm is equivalent to the shortest-job-first algorithm

D This algorithm is equivalent to the shortest-remaining-time-first algorithm

2 mins Summary

Topic Round Robin Algorithm

Topic Multilevel Queue Scheduling

Topic Multilevel Feedback Queue Scheduling

Happy Learning

THANK - YOU